

Matched-scene goal comparison

Same-scene final-frame goals: DT MAE 0.327 → 0.226. The lowest model-based matched MAE is 0.225 (DT + S3); elapsed time is 0.196. For DT + S3, matched minus separate-scene error is -0.102 (paired 95% interval [-0.124, -0.085]). Excluding the last sampled frame, DT + S3 changes 0.307 → 0.246.

DT improvement persists without the final sampled frame: MAE 0.309 → 0.233. Surprisal-only MAE gains largely disappear in that analysis. None of the full-sample DT+surprisal combinations has a paired 95% improvement interval excluding zero relative to DT alone. Elapsed time remains stronger on these uniformly successful demonstrations.

The same 12 ABC mug-flip episodes are reused (4 development, 8 test). Only the supplied visual goal changes: separate successful episode versus each episode own successful final frame. Observation codes, checkpoint, patch order, history lengths and target horizon are fixed. Calibration is refitted on development episodes only in each condition.

These are hindsight goals from successful demonstrations. Near-terminal frames resemble the target; the additional analysis drops each episode final sampled timestamp from both calibration and evaluation in both conditions. It does not remove all late successful states. This is a small paired diagnostic, not a new untouched benchmark or a failure-detection test.

Method	Separate-scene MAE	Matched MAE	Matched raw Spearman
Constant	0.328	0.328	-
DT	0.327	0.226	0.791
S1	0.332	0.301	0.403
S3	0.331	0.297	0.477
S5	0.323	0.296	0.460
Elapsed time	0.196	0.196	0.935
DT + S1	0.329	0.226	-
DT + S3	0.327	0.225	-
DT + S5	0.321	0.225	-

Without the last sampled timestamp

Method	Separate-scene MAE	Matched MAE
Constant	0.313	0.313
DT	0.309	0.233
S1	0.313	0.323
S3	0.317	0.321
S5	0.321	0.321
Elapsed time	0.191	0.191
DT + S1	0.302	0.245
DT + S3	0.307	0.246
DT + S5	0.313	0.249

All episodes finish successfully; outcome AUROC remains unavailable. Eight test episodes were already used in the earlier pilot; this is an exploratory follow-up. The earlier scene mismatch also changed goal appearance and configuration, so this experiment tests that combined change, not a specific visual cause. No DINOv2 or Robometer run is included.

protocol.json

```
{
  "seed": 20260919,
  "task": "sim turn the mug right side up",
  "checkpoint": "/ccn2b/dataset/brd/checkpoints_atlask/psi_0.5_r_abc_v2/model_00820000",
  "episodes": [
    {
      "split": "val",
      "episode_id": "episode_019e270e-c446-7199-8439-6431c74daad3",
      "frames": 709,
      "mug_qpos_addresses": [
        0,
        7
      ]
    }
  ]
}
```

```

],
"nq": 30,
"num_mugs": 2,
"final_reward": 1.0,
"ever_success": true,
"role": "dev",
"frame_indices": [
  24,
  84,
  144,
  210,
  270,
  330,
  396,
  456,
  516,
  582,
  642,
  708
],
"reward": [
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  0.0,
  0.0,
  0.0,
  0.5,
  0.5,
  0.5,
  0.5,
  0.5,
  0.5,
  1.0,
  1.0,
  1.0
],
"goal_episode_id": "episode_019e270e-c446-7199-8439-6431c74daad3",
"goal_frame": 708,
"goal_reward": 1.0
},
{
  "split": "val",
  "episode_id": "episode_019e2061-8855-7cf1-bb20-a64ffac153ec",
  "frames": 1281,
  "mug_qpos_addresses": [
    0,
    7
  ],
  "nq": 30,
  "num_mugs": 2,
  "final_reward": 1.0,
  "ever_success": true,
  "role": "dev",
  "frame_indices": [
    24,
    138,
    252,
    366,
    480,
    594,
    708,
    822,
    936,
    1050,
    1164,
    1278
  ],

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"reward": [
  0.0,
  0.0,
  0.0,
  0.0,
  0.0,
  0.0,
  0.0,
  0.5,
  0.5,
  0.5,
  1.0,
  1.0
],
"goal_episode_id": "episode_019e2061-8855-7cf1-bb20-a64ffac153ec",
"goal_frame": 1280,
"goal_reward": 1.0
},
{
  "split": "val",
  "episode_id": "episode_019e258a-7513-71b7-adff-4bfea63390e6",
  "frames": 791,
  "mug_qpos_addresses": [
    0,
    7
  ],
  "nq": 30,
  "num_mugs": 2,
  "final_reward": 1.0,
  "ever_success": true,
  "role": "dev",
  "frame_indices": [
    24,
    90,
    162,
    228,
    300,
    366,
    438,
    504,
    576,
    642,
    714,
    786
  ],
  "reward": [
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    1.0,
    1.0
  ],
  "goal_episode_id": "episode_019e258a-7513-71b7-adff-4bfea63390e6",
  "goal_frame": 790,
  "goal_reward": 1.0
},
{
  "split": "val",

```

```

"episode_id": "episode_019e3d01-3a03-706c-84e2-61080e75aa7b",
"frames": 767,
"mug_qpos_addresses": [
  0,
  7
],
"nq": 30,
"num_mugs": 2,
"final_reward": 1.0,
"ever_success": true,
"role": "dev",
"frame_indices": [
  24,
  90,
  156,
  222,
  288,
  354,
  426,
  492,
  558,
  624,
  690,
  762
],
"reward": [
  0.0,
  0.0,
  0.0,
  0.0,
  0.5,
  0.5,
  0.5,
  0.5,
  0.5,
  0.5,
  1.0,
  1.0,
  1.0
],
"goal_episode_id": "episode_019e3d01-3a03-706c-84e2-61080e75aa7b",
"goal_frame": 766,
"goal_reward": 1.0
},
{
  "split": "val",
  "episode_id": "episode_019e3c1d-21c6-7f26-a06b-d82de993e049",
  "frames": 788,
  "mug_qpos_addresses": [
    0,
    7
  ],
  "nq": 30,
  "num_mugs": 2,
  "final_reward": 1.0,
  "ever_success": true,
  "role": "test",
  "frame_indices": [
    24,
    90,
    162,
    228,
    300,
    366,
    438,
    504,

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576,
642,
714,
786
],
"reward": [
0.0,
0.0,
0.0,
0.0,
0.5,
0.5,
0.5,
0.5,
0.5,
1.0,
1.0,
1.0
],
"goal_episode_id": "episode_019e3c1d-21c6-7f26-a06b-d82de993e049",
"goal_frame": 787,
"goal_reward": 1.0
},
{
"split": "val",
"episode_id": "episode_019e3bfb-1ac2-7fd0-916c-12998d58fa5f",
"frames": 768,
"mug_qpos_addresses": [
0,
7
],
"nq": 30,
"num_mugs": 2,
"final_reward": 1.0,
"ever_success": true,
"role": "test",
"frame_indices": [
24,
90,
156,
222,
288,
354,
426,
492,
558,
624,
690,
762
],
"reward": [
0.0,
0.0,
0.0,
0.0,
0.0,
0.5,
0.5,
0.5,
0.5,
1.0,
1.0,
1.0
],
"goal_episode_id": "episode_019e3bfb-1ac2-7fd0-916c-12998d58fa5f",

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    "goal_frame": 767,
    "goal_reward": 1.0
  },
  {
    "split": "val",
    "episode_id": "episode_019e2a8d-69f8-7a1a-a3da-afbf77b1682",
    "frames": 1073,
    "mug_qpos_addresses": [
      0,
      7
    ],
    "nq": 30,
    "num_mugs": 2,
    "final_reward": 1.0,
    "ever_success": true,
    "role": "test",
    "frame_indices": [
      24,
      114,
      210,
      306,
      402,
      498,
      588,
      684,
      780,
      876,
      972,
      1068
    ],
    "reward": [
      0.0,
      0.0,
      0.0,
      0.0,
      0.0,
      0.5,
      0.5,
      0.5,
      0.5,
      0.5,
      0.5,
      1.0,
      1.0
    ],
    "goal_episode_id": "episode_019e2a8d-69f8-7a1a-a3da-afbf77b1682",
    "goal_frame": 1072,
    "goal_reward": 1.0
  },
  {
    "split": "val",
    "episode_id": "episode_019e2262-668e-7921-b3c7-e9a6cf932323",
    "frames": 651,
    "mug_qpos_addresses": [
      0,
      7
    ],
    "nq": 30,
    "num_mugs": 2,
    "final_reward": 1.0,
    "ever_success": true,
    "role": "test",
    "frame_indices": [
      24,
      78,
      132,

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192,
246,
306,
360,
420,
474,
534,
588,
648
],
"reward": [
0.0,
0.0,
0.0,
0.0,
0.5,
0.5,
0.5,
0.5,
0.5,
1.0,
1.0,
1.0
],
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"goal_frame": 650,
"goal_reward": 1.0
},
{
"split": "val",
"episode_id": "episode_019e26fc-ddf4-70ac-9383-698200cac6b3",
"frames": 663,
"mug_qpos_addresses": [
0,
7
],
"nq": 30,
"num_mugs": 2,
"final_reward": 1.0,
"ever_success": true,
"role": "test",
"frame_indices": [
24,
78,
138,
192,
252,
312,
366,
426,
486,
540,
600,
660
],
"reward": [
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0.0,
0.0,
0.0,
0.5,
0.5,
0.5,
0.5,
0.5,
0.5,
0.5,
0.5
]
}

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    1.0,
    1.0,
    1.0
  ],
  "goal_episode_id": "episode_019e26fc-ddf4-70ac-9383-698200cac6b3",
  "goal_frame": 662,
  "goal_reward": 1.0
},
{
  "split": "val",
  "episode_id": "episode_019e27bf-8328-73db-8263-d9e213c96c49",
  "frames": 623,
  "mug_qpos_addresses": [
    0,
    7
  ],
  "nq": 30,
  "num_mugs": 2,
  "final_reward": 1.0,
  "ever_success": true,
  "role": "test",
  "frame_indices": [
    24,
    78,
    132,
    186,
    240,
    294,
    348,
    402,
    456,
    510,
    564,
    618
  ],
  "reward": [
    0.0,
    0.0,
    0.0,
    0.0,
    0.0,
    0.5,
    0.5,
    0.5,
    0.5,
    1.0,
    1.0,
    1.0
  ],
  "goal_episode_id": "episode_019e27bf-8328-73db-8263-d9e213c96c49",
  "goal_frame": 622,
  "goal_reward": 1.0
},
{
  "split": "val",
  "episode_id": "episode_019e1d6a-4038-7013-bffe-e69df784231a",
  "frames": 616,
  "mug_qpos_addresses": [
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    7
  ],
  "nq": 30,
  "num_mugs": 2,
  "final_reward": 1.0,
  "ever_success": true,

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"role": "test",
"frame_indices": [
  24,
  72,
  126,
  180,
  234,
  288,
  342,
  396,
  450,
  504,
  558,
  612
],
"reward": [
  0.0,
  0.0,
  0.0,
  0.0,
  0.5,
  0.5,
  0.5,
  0.5,
  0.5,
  0.5,
  1.0,
  1.0,
  1.0
],
"goal_episode_id": "episode_019e1d6a-4038-7013-bffe-e69df784231a",
"goal_frame": 615,
"goal_reward": 1.0
},
{
  "split": "val",
  "episode_id": "episode_019e2891-b03f-7e80-a987-2e2e8431a80e",
  "frames": 703,
  "mug_qpos_addresses": [
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    7
  ],
  "nq": 30,
  "num_mugs": 2,
  "final_reward": 1.0,
  "ever_success": true,
  "role": "test",
  "frame_indices": [
    24,
    84,
    144,
    204,
    270,
    330,
    390,
    450,
    516,
    576,
    636,
    702
  ],
  "reward": [
    0.0,
    0.0,
    0.0,
    0.0,

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    0.0,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    1.0,
    1.0
  ],
  "goal_episode_id": "episode_019e2891-b03f-7e80-a987-2e2e8431a80e",
  "goal_frame": 702,
  "goal_reward": 1.0
}
],
"histories": [
  1,
  3,
  5
],
"patches_per_frame": 128,
"target_offset_slots": 5,
"fps": 5,
"image_hw": [
  256,
  336
],
"label_rule": "fraction of mugs upright_z >= .95 and height <= first_frame_height + .03 m",
"scope": "Paired diagnostic: same 12 episodes, splits, 144 timestamps, observation codes, patch order and checkpoint; change only reference goal to each successful episode final frame.",
"goal_mode": "matched_final_frame",
"previous_independent_goal": {
  "split": "val",
  "episode_id": "episode_019e3937-3fda-7f73-a7ea-3333280a53a1",
  "frames": 809,
  "mug_qpos_addresses": [
    0,
    7
  ],
  "nq": 30,
  "num_mugs": 2,
  "final_reward": 1.0,
  "ever_success": true,
  "role": "goal",
  "frame_indices": [
    808
  ]
},
"endpoint_check": "Additionally exclude the final sampled timestamp from every episode in both conditions, including development calibration; compare 132 common timestamps."
}

```

paired_comparison.json

```

{
  "all_timestamps": {
    "Constant": {
      "independent_mae": 0.328125,
      "matched_mae": 0.328125,
      "matched_minus_independent": 0.0,
      "paired_95ci": [
        0.0,
        0.0
      ],
      "matched_raw_spearman": null
    },
    "DT": {

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"independent_mae": 0.3273405337038059,
"matched_mae": 0.22605856672076002,
"matched_minus_independent": -0.10128196698304591,
"paired_95ci": [
  -0.12625635664920232,
  -0.07936727509181293
],
"matched_raw_spearman": 0.7913272497210598
},
"S1": {
  "independent_mae": 0.3319016533156805,
  "matched_mae": 0.3008453747210814,
  "matched_minus_independent": -0.031056278594599127,
  "paired_95ci": [
    -0.04004511676793792,
    -0.02298508227781136
  ],
  "matched_raw_spearman": 0.4031861135364723
},
"S3": {
  "independent_mae": 0.3310623249560374,
  "matched_mae": 0.2968709329323668,
  "matched_minus_independent": -0.03419139202367055,
  "paired_95ci": [
    -0.0481253492798148,
    -0.019443233753519158
  ],
  "matched_raw_spearman": 0.4770141560575599
},
"S5": {
  "independent_mae": 0.3234628925934101,
  "matched_mae": 0.2960012566009038,
  "matched_minus_independent": -0.027461635992506266,
  "paired_95ci": [
    -0.047920657243091976,
    -0.009869112498078191
  ],
  "matched_raw_spearman": 0.4601199293099806
},
"Elapsed time": {
  "independent_mae": 0.19623382293657682,
  "matched_mae": 0.19623382293657682,
  "matched_minus_independent": 0.0,
  "paired_95ci": [
    0.0,
    0.0
  ],
  "matched_raw_spearman": 0.9353970709613129
},
"DT + S1": {
  "independent_mae": 0.32875687786060204,
  "matched_mae": 0.22584291248846752,
  "matched_minus_independent": -0.10291396537213456,
  "paired_95ci": [
    -0.12377656692738194,
    -0.08512264518090534
  ],
  "matched_raw_spearman": null
},
"DT + S3": {
  "independent_mae": 0.3271711423834641,
  "matched_mae": 0.2249612402906484,
  "matched_minus_independent": -0.10220990209281566,
  "paired_95ci": [
    -0.12408693979632204,

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    -0.0847175825619119
  ],
  "matched_raw_spearman": null
},
"DT + S5": {
  "independent_mae": 0.32063680103357006,
  "matched_mae": 0.22535581035956148,
  "matched_minus_independent": -0.09528099067400855,
  "paired_95ci": [
    -0.12387060537590008,
    -0.07153092689976756
  ],
  "matched_raw_spearman": null
}
},
"without_final_sample": {
  "Constant": {
    "independent_mae": 0.3130165289256198,
    "matched_mae": 0.3130165289256198,
    "matched_minus_independent": 0.0,
    "paired_95ci": [
      0.0,
      0.0
    ],
    "matched_raw_spearman": null
  },
  "DT": {
    "independent_mae": 0.30921456038846373,
    "matched_mae": 0.23327273306199778,
    "matched_minus_independent": -0.07594182732646598,
    "paired_95ci": [
      -0.09640572385821185,
      -0.058588394489001154
    ],
    "matched_raw_spearman": 0.733418222368106
  },
  "S1": {
    "independent_mae": 0.313150563270962,
    "matched_mae": 0.3228924904438456,
    "matched_minus_independent": 0.00974192717288358,
    "paired_95ci": [
      0.006674610137514313,
      0.012894141731482612
    ],
    "matched_raw_spearman": 0.23412470357057713
  },
  "S3": {
    "independent_mae": 0.31695521834017126,
    "matched_mae": 0.3210536648260821,
    "matched_minus_independent": 0.0040984464859109004,
    "paired_95ci": [
      -0.000446463689708324,
      0.007810186650685514
    ],
    "matched_raw_spearman": 0.332645547686473
  },
  "S5": {
    "independent_mae": 0.3211957975236268,
    "matched_mae": 0.32122867435318214,
    "matched_minus_independent": 3.287682955534221e-05,
    "paired_95ci": [
      -0.004038547627877684,
      0.004618834887753284
    ],
    "matched_raw_spearman": 0.310887420521725
  }
}

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    },
    "Elapsed time": {
      "independent_mae": 0.19092968314140077,
      "matched_mae": 0.19092968314140077,
      "matched_minus_independent": 0.0,
      "paired_95ci": [
        0.0,
        0.0
      ],
      "matched_raw_spearman": 0.9212871866683131
    },
    "DT + S1": {
      "independent_mae": 0.3020279566611934,
      "matched_mae": 0.24450486020939394,
      "matched_minus_independent": -0.05752309645179947,
      "paired_95ci": [
        -0.07811648692983915,
        -0.04009111453777915
      ],
      "matched_raw_spearman": null
    },
    "DT + S3": {
      "independent_mae": 0.3074678910396515,
      "matched_mae": 0.2463939179635804,
      "matched_minus_independent": -0.0610739730760711,
      "paired_95ci": [
        -0.08029365706237573,
        -0.04472649508291952
      ],
      "matched_raw_spearman": null
    },
    "DT + S5": {
      "independent_mae": 0.31305887153241546,
      "matched_mae": 0.2485948634525153,
      "matched_minus_independent": -0.06446400807990016,
      "paired_95ci": [
        -0.08091985716950918,
        -0.050187938334665415
      ],
      "matched_raw_spearman": null
    }
  }
}

```

metrics.json

```

{
  "checkpoint_step": 820000,
  "task": "mug flipping",
  "dev_episodes": 4,
  "test_episodes": 8,
  "timestamps_per_episode": 12,
  "goal_mode": "matched final frame",
  "exclude_final_sample": false,
  "methods": {
    "Constant": {
      "test_episode_macro_mae": 0.328125,
      "mae_95ci": [
        0.31857638888888884,
        0.33854166666666667
      ],
      "test_episode_macro_spearman": null,
      "valid_correlation_episodes": 0,
      "raw_score_spearman": null,
      "features": [],
      "standardization_mean": [],

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```

"standardization_std": [],
"coefficients": []
},
"DT": {
  "test_episode_macro_mae": 0.22605856672076002,
  "mae_95ci": [
    0.21004258454702446,
    0.24241753968105004
  ],
  "test_episode_macro_spearman": 0.787075810128691,
  "valid_correlation_episodes": 8,
  "raw_score_spearman": 0.7913272497210598,
  "features": [
    "DT"
  ],
  "standardization_mean": [
    -68.64767620712519
  ],
  "standardization_std": [
    51.44173406028894
  ],
  "coefficients": [
    0.3958333333333333,
    0.22353910478873532
  ]
},
"S1": {
  "test_episode_macro_mae": 0.3008453747210814,
  "mae_95ci": [
    0.28727212814737235,
    0.3123603443855311
  ],
  "test_episode_macro_spearman": 0.4031861135364723,
  "valid_correlation_episodes": 8,
  "raw_score_spearman": 0.4031861135364723,
  "features": [
    "S1"
  ],
  "standardization_mean": [
    -7.596841479030748
  ],
  "standardization_std": [
    2.102066231191038
  ],
  "coefficients": [
    0.3958333333333333,
    0.18670948565570092
  ]
},
"S3": {
  "test_episode_macro_mae": 0.2968709329323668,
  "mae_95ci": [
    0.2812260909085169,
    0.3098416159836577
  ],
  "test_episode_macro_spearman": 0.4770141560575599,
  "valid_correlation_episodes": 8,
  "raw_score_spearman": 0.4770141560575599,
  "features": [
    "S3"
  ],
  "standardization_mean": [
    -7.416162585218747
  ],
  "standardization_std": [

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    1.9899154448856193
  ],
  "coefficients": [
    0.3958333333333333,
    0.1932362767576606
  ]
},
"S5": {
  "test_episode_macro_mae": 0.2960012566009038,
  "mae_95ci": [
    0.28042948073304313,
    0.30953100222801
  ],
  "test_episode_macro_spearman": 0.4601199293099806,
  "valid_correlation_episodes": 8,
  "raw_score_spearman": 0.4601199293099806,
  "features": [
    "S5"
  ],
  "standardization_mean": [
    -7.2471819159885245
  ],
  "standardization_std": [
    1.942417191122498
  ],
  "coefficients": [
    0.3958333333333333,
    0.1970489289365981
  ]
},
"Elapsed time": {
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  "mae_95ci": [
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    0.2185598614881379
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  "test_episode_macro_spearman": 0.9353970709613129,
  "valid_correlation_episodes": 8,
  "raw_score_spearman": 0.9353970709613129,
  "features": [
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  "standardization_mean": [
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  "standardization_std": [
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  "coefficients": [
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    0.27525928339574546
  ]
},
"DT + S1": {
  "test_episode_macro_mae": 0.22584291248846752,
  "mae_95ci": [
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    0.24272284083405946
  ],
  "test_episode_macro_spearman": 0.7620091050265163,
  "valid_correlation_episodes": 8,
  "raw_score_spearman": null,
  "features": [
    "DT",
    "S1"
  ]
}

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],
"standardization_mean": [
  -68.64767620712519,
  -7.596841479030748
],
"standardization_std": [
  51.44173406028893,
  2.102066231191038
],
"coefficients": [
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  0.1774045368724406,
  0.11639007120522918
]
},
"DT + S3": {
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  "mae_95ci": [
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    0.24207211676276216
  ],
  "test_episode_macro_spearman": 0.7698585077666522,
  "valid_correlation_episodes": 8,
  "raw_score_spearman": null,
  "features": [
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    "S3"
  ],
  "standardization_mean": [
    -68.64767620712519,
    -7.416162585218747
  ],
  "standardization_std": [
    51.44173406028893,
    1.9899154448856193
  ],
  "coefficients": [
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    0.1227577751707836
  ]
},
"DT + S5": {
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  "mae_95ci": [
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    0.24170009453482838
  ],
  "test_episode_macro_spearman": 0.7520330515268486,
  "valid_correlation_episodes": 8,
  "raw_score_spearman": null,
  "features": [
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    "S5"
  ],
  "standardization_mean": [
    -68.64767620712519,
    -7.2471819159885245
  ],
  "standardization_std": [
    51.44173406028893,
    1.9424171911224983
  ],
  "coefficients": [
    0.3958333333333333,

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    0.17106186829150197,
    0.12554697324845093
  ]
}
},
"paired_comparisons": {
  "S1": {
    "combo_minus_DT_mae": -0.0002156542322924726,
    "paired_episode_bootstrap_95ci": [
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      0.006057783491093704
    ]
  },
  "S3": {
    "combo_minus_DT_mae": -0.001097326430111574,
    "paired_episode_bootstrap_95ci": [
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      0.005424652301094084
    ]
  },
  "S5": {
    "combo_minus_DT_mae": -0.000702756361198504,
    "paired_episode_bootstrap_95ci": [
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      0.0060250174380202795
    ]
  }
},
"calibration": "z-score on development frames; ridge alpha=1, unpenalized intercept; clip predictions
to [0,1]; no hyperparameter selection",
"terminal_outcome_aucroc": null,
"terminal_outcome_reason": "12/12 evaluation episodes successful; no failed terminal examples",
"dt_diagnostics": {
  "dt_expected_bins": {
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    "max": 320.56256103515625,
    "mean": 75.40430176589224
  },
  "dt_entropy": {
    "min": 0.5445351600646973,
    "max": 6.300136566162109,
    "mean": 4.3862472296588955
  },
  "dt_valid_mass": {
    "min": 0.6039844155311584,
    "max": 0.9999751448631287,
    "mean": 0.9068699181079865
  },
  "dt_last_bin_mass": {
    "min": 1.3766284986949096e-11,
    "max": 4.3234385884716175e-06,
    "mean": 6.48132079213692e-07
  }
},
"limitations": [
  "One task; eight test episodes; all successful demonstrations.",
  "Hindsight final-frame goal from the same successful episode; endpoint appearance can make this
diagnostic easier than a prospectively supplied goal.",
  "128 fixed shuffled patches/image; controlled history comparison differs from the training uniform
total-budget allocation.",
  "Ridge fits can reverse raw-score signs; raw correlation and coefficients are reported.",
  "Training membership not fully audited; no zero-shot generalization claim.",
  "DINOv2, Robometer, bottles and plates have not been run."
]
}

```

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pairing_validation.json
{
  "result": "passed",
  "paired_episodes": 12,
  "paired_timestamps": 144,
  "observation_codes": "byte-identical cached observations",
  "goals": "each episode own final frame, verified completion=1",
  "base_protocol_sha256": "a28afceec403f903db798126cb602777a8154d9bdfd362d3942c1edda4387da7",
  "matched_protocol_sha256": "9adc5dfd72182a77ace4307f6bc01c61c5e7fca021b900456239a9e808c30ab2"
}
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